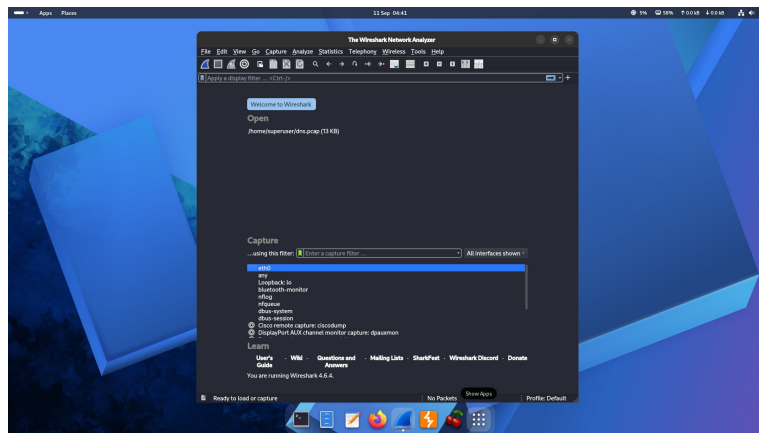


Wireshark Traffic Analysis Report

Wireshark Traffic Analysis — Report

1. Setup

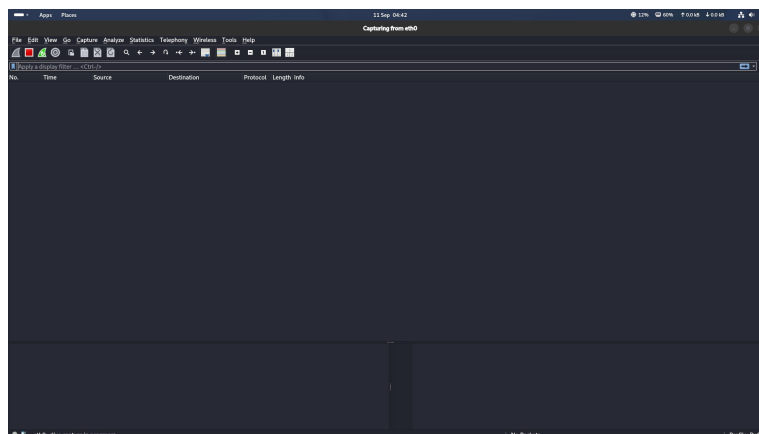
Wireshark 4.6.4 was launched on a Kali Linux VM. The `eth0` interface was selected as the capture target since it showed live traffic activity.



Wireshark launch screen

2. Starting the capture

A live capture was started on `eth0`. At this point no packets had been captured yet.



Capture started, no packets

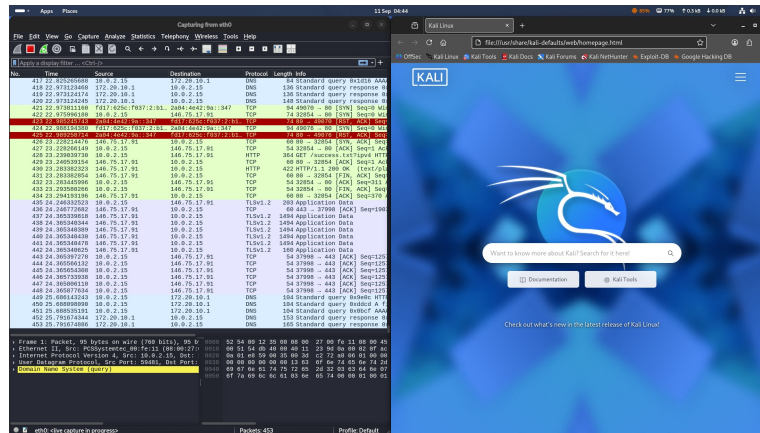
3. Generating traffic

A web browser was launched to generate realistic traffic. Within seconds, the capture picked up:

- **DNS** queries and responses (e.g., resolving

firefox.settings.services.mozilla.com)

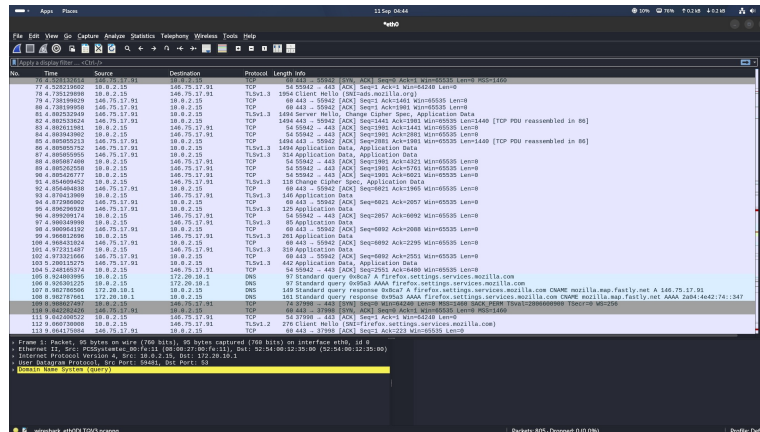
- **TCP** handshakes (SYN / SYN-ACK / ACK) to remote hosts
- A **plaintext HTTP** request/response — GET /success.txt?ipv4 — a captive-portal / connectivity check request sent with no encryption
- **TLS 1.2 / TLS 1.3** encrypted application traffic



Browser launch, capture in progress

4. Full capture review

By the end of the capture (805 packets), the traffic included TLS 1.3 handshakes with visible **SNI (Server Name Indication)** values such as ads.mozilla.org, plus DNS request/response pairs resolving domains to CNAMES and A/AAAA records.



Full capture with TLS/DNS

5. Findings

Traffic type	Observation	Security implication
DNS	Queries/responses sent in cleartext (UDP/53)	Anyone on the network path can see every domain a device resolves
HTTP	Full GET request and 200 OK response visible in plaintext	Any data in an HTTP request/response (headers, query params, body) is readable by a sniffer
TLS 1.2/1.3	Application data is	Payload confidentiality is

	encrypted	preserved, but...
TLS Client Hello (SNI)	Destination hostname sent in cleartext during the handshake	Even encrypted connections leak <i>which domains</i> a device is talking to

6. Conclusion

This capture demonstrates why HTTPS-everywhere matters: even a short browsing session leaks domain-level metadata via DNS and TLS SNI, and any remaining plaintext HTTP traffic is fully readable by anyone capturing packets on the same network segment.

Recommended mitigations include enforcing HTTPS, using DNS-over-HTTPS/TLS, and considering Encrypted Client Hello (ECH) where supported, to reduce metadata leakage.